

LECTURE 9 LINEAR PROCESSES I: WOLD DECOMPOSITION

In this lecture, we focus on covariance stationary processes.

Definition 1 (*White noise*) A process $\{\varepsilon_t\}$ is called white noise (WN) if $E[\varepsilon_t] = 0$, $E[\varepsilon_t^2] = \sigma^2 < \infty$ and $E[\varepsilon_t \varepsilon_{t-j}] = 0$ for all t and $j \neq 0$.

Definition 2 (*Moving average*) A process $\{u_t\}$ is called a moving average process of order q (MA(q)) if

$$u_t = c_0 \varepsilon_t + c_1 \varepsilon_{t-1} + \dots + c_q \varepsilon_{t-q}, \quad (1)$$

and $\{\varepsilon_t\}$ is a WN.

A process of the form (1) is called linear. The Wold decomposition says that any mean zero covariance stationary process with absolutely summable autocovariances can be represented in the MA(∞) form.

For a covariance stationary process, we assume that second moments are finite. Let L_2 denote the space of random variables with finite second moments. For $X, Y \in L_2$ define the inner product

$$\langle X, Y \rangle = E[XY].$$

When equipped with such a definition of the inner product, L_2 is a Hilbert space. Consider the mean zero covariance stationary process $\{X_t\}$, such that

$$\sum_{j=0}^{\infty} |\gamma(j)| < \infty, \quad (2)$$

where

$$\gamma(j) = E[X_t X_{t-j}].$$

Define $\mathcal{M}_t$ to be the smallest closed subspace of L_2 that contains all elements of the form

$$\sum_{j=0}^{\infty} c_j X_{t-j} \text{ such that } \sum_{j=0}^{\infty} c_j^2 < \infty.$$

The requirement $\sum_{j=0}^{\infty} c_j^2 < \infty$ is to ensure that the elements of $\mathcal{M}_t$ are in L_2 . Indeed, in this case, the variance of any element of $\mathcal{M}_t$ can be bounded using $\sum_{j=0}^{\infty} c_j^2$ and $\sum_{j=0}^{\infty} |\gamma(j)|$:

$$\begin{aligned} \text{Var} \left(\sum_{j=0}^{\infty} c_j X_{t-j} \right) &= \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} c_i c_j \gamma(i-j) \\ &\leq 2 \sum_{h=0}^{\infty} \sum_{j=0}^{\infty} c_j c_{j+h} \gamma(h) \\ &= 2 \sum_{h=0}^{\infty} \gamma(h) \sum_{j=0}^{\infty} c_j c_{j+h} \\ &\leq 2 \sum_{h=0}^{\infty} |\gamma(h)| \left| \sum_{j=0}^{\infty} c_j c_{j+h} \right| \\ &\leq 2 \sum_{h=0}^{\infty} |\gamma(h)| \left(\sum_{j=0}^{\infty} c_j^2 \right)^{1/2} \left(\sum_{j=0}^{\infty} c_{j+h}^2 \right)^{1/2} \\ &\leq 2 \left(\sum_{j=0}^{\infty} c_j^2 \right) \sum_{h=0}^{\infty} |\gamma(h)| \\ &< \infty, \end{aligned}$$

provided that (2) holds. Furthermore, $\mathcal{M}_t$ is an increasing sequence:

$$\dots \subset \mathcal{M}_t \subset \mathcal{M}_{t+1} \subset \dots$$

Let $P_{\mathcal{M}_t}$ be the orthogonal projection onto $\mathcal{M}_t$. We can write

$$X_t = \widehat{X}_t + \varepsilon_t, \quad (3)$$

where

$$\begin{aligned} \widehat{X}_t &= P_{\mathcal{M}_{t-1}} X_t, \\ \varepsilon_t &= (1 - P_{\mathcal{M}_{t-1}}) X_t. \end{aligned}$$

By the Hilbert space projection theorem, $\widehat{X}_t$ solves the least squares problem. Therefore, $\widehat{X}_t$ can be interpreted as the best one-step-ahead linear predictor of X_t (in the mean squared error sense), and ε_t is the prediction error. We have $\widehat{X}_t \in \mathcal{M}_{t-1}$, and $\varepsilon_t \in \mathcal{M}_t$, since

$$\varepsilon_t = X_t - \widehat{X}_t.$$

Furthermore, by the orthogonal projection result

$$\varepsilon_t \in \mathcal{M}_{t-1}^\perp,$$

where

$$\mathcal{M}_t^\perp = \{Y \in L_2 : \langle Y, X \rangle = 0 \text{ for all } X \in \mathcal{M}_t\}.$$

Since $\mathcal{M}_t$ is an increasing sequence, it includes the members of $\mathcal{M}_{t-h}$ and we have that $\varepsilon_t \in \mathcal{M}_{t-h}^\perp$ for all $h \geq 1$. Therefore

$$\mathbb{E}[\varepsilon_t \varepsilon_{t-h}] = 0 \text{ for all } h \geq 1.$$

Furthermore,

$$\mathbb{E}[\varepsilon_t^2] = \sigma^2 < \infty$$

and constant for all t since $\{X_t\}$ is covariance stationary.

Define

$$\mathcal{E}_t = \left\{ \sum_{j=0}^{\infty} c_j \varepsilon_{t-j} : \sum_{j=0}^{\infty} c_j^2 < \infty \right\},$$

where $\mathcal{E}_t$ is actually a closed linear subspace of L_2 . Let $P_{\mathcal{E}_t}$ be the orthogonal projection onto $\mathcal{E}_t$. We have

$$X_t = P_{\mathcal{E}_t} X_t + V_t, \quad (4)$$

where, since $\mathcal{E}_t$ is closed and linear,

$$P_{\mathcal{E}_t} X_t = \sum_{j=0}^{\infty} \widehat{c}_j \varepsilon_{t-j} \quad (5)$$

for some sequence $\{\widehat{c}_j\}$, and

$$V_t = (1 - P_{\mathcal{E}_t}) X_t.$$

Since $V_t \in \mathcal{M}_t$, and because of (3),

$$\mathcal{M}_t = \mathcal{M}_{t-1} \oplus \mathcal{S}(\varepsilon_t),$$

where $\mathcal{S}(\varepsilon_t)$ is the linear subspace spanned by ε_t , and $\oplus$ denotes the direct sum:

$$\mathcal{S}_1 \oplus \mathcal{S}_2 = \{x_1 + x_2 : x_1 \in \mathcal{S}_1, x_2 \in \mathcal{S}_2\}.$$

Since P_{ε_t} is the orthogonal projection, $E[V_t \varepsilon_t] = 0$ by (4) and (5), and therefore $V_t \notin \mathcal{S}(\varepsilon_t)$. Therefore, it must be true that $V_t \in \mathcal{M}_{t-1}$. By the same argument, since

$$\begin{aligned}\mathcal{M}_{t-1} &= \mathcal{M}_{t-2} \oplus \mathcal{S}(\varepsilon_{t-1}), \text{ and} \\ E[V_t \varepsilon_{t-1}] &= 0,\end{aligned}$$

we deduce that V_t belongs to each of $\mathcal{M}_{t-1}, \mathcal{M}_{t-2}, \dots$. Let

$$\mathcal{M}_{-\infty} = \bigcap_{t=-\infty}^{\infty} \mathcal{M}_t.$$

We conclude that

$$V_t \in \mathcal{M}_{-\infty} \text{ for all } t.$$

Thus, V_t is an element of any linear subspace $\mathcal{M}_s$, $s \in \mathbb{Z}$, where $\mathbb{Z}$ is the set of integers. The entire process $\{V_t\}$ can be predicted with certainty from an arbitrary distant past of $\{X_t\}$. Such a process is called deterministic.

We derived the Wold representation for a covariance stationary process $\{X_t\}$:

$$X_t = \sum_{j=0}^{\infty} c_j \varepsilon_{t-j} + V_t,$$

where $\{\varepsilon_t\}$ is a WN, and V_t is deterministic. When $V_t = 0$ for all t , the process $\{X_t\}$ is said to be linearly indeterministic. If $\{X_t\}$ is indeterministic and mean zero, it has the MA(∞) representation

$$X_t = \sum_{j=0}^{\infty} c_j \varepsilon_{t-j}.$$

It follows from the projection properties that

$$c_j = E[X_t \varepsilon_{t-j}] / \sigma^2, \text{ and} \tag{6}$$

$$c_0 = 1. \tag{7}$$

Remarks:

1. The representation is unique with probability one, which follows from uniqueness of the Hilbert space projection.
2. The WN elements in the Wold decomposition are the one-step-ahead linear prediction errors.
3. Sometimes ε_t 's are interpreted as the fundamental shocks of the economy. Then the impulse responses, c_j 's, represent the effect of a shock after j periods. Suppose that the fundamental shocks are given by the unexpected shocks to the agents' information set. Let $\mathcal{F}_t = \sigma(X_t, X_{t-1}, \dots)$. $\mathcal{F}_t$ represents the information of the agents in the economy at time t . The predictor $\hat{X}_t$ is restricted to be a linear predictor, and, therefore, is not necessarily equal to $E[X_t | \mathcal{F}_t]$. Hence, the true fundamental shocks may differ from the shocks that enter the Wold representation (see, for example, Hansen and Sargent (1991), Chapter 4).